

MULTIPLE REGRESSION WITH AUTOCORRELATED ERRORS: BAYESIAN ANALYSIS WITH DIFFERENT PRIORS

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Summary: Known results for the Bayesian analysis of the multiple regression model with AR(1) errors are extended and expressions obtained for the predictive density of a future observation for certain prior information. The influence of prior information on the predictive credibility interval is investigated through a simulation study.

1. INTRODUCTION

A model generally associated with time series data is the multiple regression model given by

$$(1) \quad y = X\beta + \epsilon,$$

where y is an $n \times 1$ vector of values of the response variable, X is an $n \times p$ matrix of known predictors, β is a $p \times 1$ vector of unknown regression parameters and ϵ is an $n \times 1$ vector of random errors with $E[\epsilon] = 0$ and $E[\epsilon\epsilon'] = \Phi = \sigma^2\Psi$, where Ψ is given by

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$$\Psi = \frac{1}{1 - \rho^2} \begin{bmatrix} 1 & \rho & \rho^2 & \dots & \rho^{n-1} \\ \rho & 1 & \rho & \dots & \rho^{n-2} \\ \rho^2 & \rho & 1 & \dots & \rho^{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \rho^{n-1} & \rho^{n-2} & \rho^{n-3} & \dots & 1 \end{bmatrix}$$

Bayesian analysis of this model was initiated by Zellner and Tiao (1964), when they considered a multiple regression model with errors generated by an AR(1) process, and assumed non-informative priors for the parameters of the model. This work was extended by Richard (1977), who considered a general k -th order autoregressive process for the errors, introduced normal-gamma and Student-gamma priors into the analysis, but derived all densities conditional on the parameters of the autoregressive process. Richard (1977) also obtained the predictive density of y but not the predictive density of future observations conditional on y .

Broemeling (1985:207) has considered the autocorrelation model with a single predictor variable, an AR(1) process for the errors and informative priors for all the parameters except for an uninformative prior for ρ , and discusses methods for obtaining the moments of the posterior densities. Broemeling (1985:401) has also considered multiple regression with autocorrelated errors and used uninformative priors for all parameters.

We follow the approach of Judge, Griffiths, Hill, R.C., Lütkepohl and Lee (1985:291), and consider the transformed model

$$(2) \quad y^* = X^* \beta + e^*,$$

where $y^* = Py$, $X^* = PX$, $e^* = Pe$, and $P'P = \Psi^{-1}$

Judge, *et al.* (1985:291) have obtained a Bayesian estimate of β with model (1) and AR(1) errors, using a non-informative prior for (β, ρ, σ) . The following sections provide an extension of these results to those obtained when a normal-gamma prior is assumed for (β, σ) , and the predictive

density of a future observation in each case is obtained. These predictive densities are of particular importance in the time series context, where the general aim is forecasting, and they can provide a credibility interval for future observations. Since these densities are not in analytic form, they require numerical integration.

2. NON-INFORMATIVE PRIOR

These results are given by Judge, et al. (1985:291).

With a non-informative prior for (β, ρ, σ) , the joint posterior density is

$$f(\beta, \rho, \sigma | y) \propto \sigma^{-(n+1)} \exp \left\{ -\frac{1}{2\sigma^2} (y^* - X^* \beta)' (y^* - X^* \beta) \right\}.$$

Integrating out σ , the joint posterior density for β and ρ is

$$f(\beta, \rho | y) \propto (RSS)^{-1} \left[1 + \frac{(\beta - \hat{\beta})' X^* X^* (\beta - \hat{\beta})}{RSS} \right]^{-\frac{n}{2}},$$

where $\hat{\beta} = (X^{*'} X^*)^{-1} X^{*'} y^*$ and $RSS = (y^* - X^* \hat{\beta})' (y^* - X^* \hat{\beta})$.

To obtain the Bayesian estimator, $E[\beta | y]$, the density $f(\rho | y)$ is first obtained by integrating out β , to give

$$f(\rho | y) \propto (RSS)^{-\frac{(n-p)}{2}} |X^{*'} X^*|^{-\frac{1}{2}}.$$

This density is then used in

$$E[\beta | y] = \int \int \beta f(\beta, \rho | y) d\beta d\rho = \int \int \beta f(\beta | \rho, y) f(\rho | y) d\beta d\rho = \int \hat{\beta} f(\rho | y) d\rho.$$

Firstly, numerical integration is used to find the normalising constant for $f(\rho | y)$ and again to obtain $E[\beta | y]$.

3. NORMAL-GAMMA PRIOR

With the assumption that β has a prior multivariate normal density, and σ has a prior inverted gamma density, the joint posterior density is given by

$$f(\beta, \rho, \sigma | y) \propto \sigma^{-(n+p+1)} \exp \left\{ -\frac{1}{2\sigma^2} [y^2 + (y^* - X^* \beta)' (y^* - X^* \beta) + (\beta - \bar{\beta})' A (\beta - \bar{\beta})] \right\},$$

where β , s^2 , v and A depend on the prior information.

After integrating out σ , the density becomes

$$\begin{aligned} f(\beta, \rho | y) &\propto [\rho s^2 + (y^* - X^* \beta)'(y^* - X^* \beta) + (\beta - \bar{\beta})' A (\beta - \bar{\beta})]^{-\frac{(n+p+v)}{2}} \\ &\propto [\rho s^2 + y^{*'} y^* + \bar{\beta}' A \bar{\beta} - (A \bar{\beta} + X^{*'} y^*)' (X^{*'} X^* + A)^{-1} (A \bar{\beta} + X^{*'} y^*) \\ &\quad + (\beta - (X^{*'} X^* + A)^{-1} (A \bar{\beta} + X^{*'} y^*))' (X^{*'} X^* + A)^{-1} \\ &\quad \times (\beta - (X^{*'} X^* + A)^{-1} (A \bar{\beta} + X^{*'} y^*))]^{-\frac{(n+p+v)}{2}} \end{aligned}$$

This is the kernel of a multivariate t -distribution, so that if ρ is known, as assumed by Richard (1977), the Bayesian estimate of β would be $(X^{*'} X^* + A)^{-1} (A \bar{\beta} + X^{*'} y^*)$. By integrating out β , the posterior density of ρ is given by

$$\begin{aligned} f(\rho | y) &\propto [\rho s^2 + y^{*'} y^* + \bar{\beta}' A \bar{\beta} - (A \bar{\beta} + X^{*'} y^*)' (X^{*'} X^* + A)^{-1} \\ &\quad \times (A \bar{\beta} + X^{*'} y^*)]^{-\frac{(n+p+v)}{2}} |X^{*'} X^* + A|^{\frac{1}{2}} \end{aligned}$$

Numerical integration is used to find the normalising constant and then $f(\rho | y)$ is used to find the Bayesian estimate of β using

$$E[\beta | y] = \int \beta f(\rho | y) d\rho$$

4. PREDICTIVE DENSITY OF A FUTURE OBSERVATION

4.1 Introduction

For the general linear model with AR(1) errors we have

$$\begin{aligned} y_i &= x_i' \beta + e_i \\ e_i &= \rho e_{i-1} + v_i \end{aligned}$$

If we consider the same transformed model as in (2) then

$$y_1^* = X_1^* \beta + e_1^*$$

where the subscript 1 indicates the sample data.

We wish to predict the $(n+1)$ -th observation,

$$\tilde{y} = \tilde{x}' \beta + e_{n+1} = \tilde{x}' \beta + \rho e_n + v_{n+1}$$

We can now consider the transformed model including $\tilde{y}$, i.e.

$$y^* = X^* \beta + e^*$$

where

$$y^* = \begin{bmatrix} y_1^* \\ \tilde{y} - \rho y_n \end{bmatrix} \text{ and } X^* = \begin{bmatrix} X_1^* \\ \tilde{x}' - \rho x_n' \end{bmatrix}$$

The matrix P is as before but is now of order $n+1$.

The joint likelihood is

$$\begin{aligned} L(\rho, \beta, \sigma) &\propto (1 - \rho^2)^{\frac{1}{2}} \sigma^{-(n+1)} \exp \left\{ -\frac{1}{2\sigma^2} (y^* - X^* \beta)' (y^* - X^* \beta) \right\} \\ &\propto (1 - \rho^2)^{\frac{1}{2}} \sigma^{-(n+1)} \exp \left\{ -\frac{1}{2\sigma^2} [(y_1^* - X_1^* \beta)' (y_1^* - X_1^* \beta) + (\tilde{y} - \rho y_n - (\tilde{x} - \rho x_n)' \beta)^2] \right\} \end{aligned}$$

4.2 Non-informative prior

Assuming a non-informative prior, i.e.

$$(3) \quad f(\beta, \rho, \sigma) \propto (1 - \rho^2)^{-\frac{1}{2}} \sigma^{-1}$$

the joint posterior density is

$$f(\tilde{y}, \beta, \rho, \sigma | y) \propto \sigma^{-(n+2)} \exp \left\{ -\frac{1}{2\sigma^2} [(y_1^* - X_1^* \beta)' (y_1^* - X_1^* \beta) + (\tilde{y} - \rho y_n - (\tilde{x} - \rho x_n)' \beta)^2] \right\}$$

On integrating out σ , the joint posterior density for $\tilde{y}$, β and ρ is

$$\begin{aligned} f(\tilde{y}, \beta, \rho | y) &\propto [(y_1^* - X_1^* \beta)' (y_1^* - X_1^* \beta) + (\tilde{y} - \rho y_n - (\tilde{x} - \rho x_n)' \beta)^2]^{-\left(\frac{n+1}{2}\right)} \\ &\propto [(y^* - X^* \beta)' (y^* - X^* \beta) + (\beta - \hat{\beta})' X^* X^* (\beta - \hat{\beta})]^{-\left(\frac{n+1}{2}\right)} \end{aligned}$$

where $\hat{\beta} = (X^{*'} X^*)^{-1} X^{*'} y^*$.

On integrating out β , we obtain

$$f(\bar{y}, \rho | y) \propto |X^*{}' X^*|^{-\frac{1}{2}} [(y^* - X^* \beta)' (y^* - X^* \beta)]^{-\frac{(n+1-p)}{2}}$$

where $y^*{}' = [y_1^* (\bar{y} - \rho y_n)]'$, and $X^*{}' = [X_1^* (\bar{x} - \rho x_n)]'$.

Now

$$\begin{aligned} (y^* - X^* \beta)' (y^* - X^* \beta) &= y^*{}' y^* - y^*{}' X^* (X^*{}' X^*)^{-1} X^*{}' y^* \\ &= y_1^*{}' y_1^* + (\bar{y} - \rho y_n)^2 - [y_1^*{}' X_1^* + (\bar{y} - \rho y_n)(\bar{x} - \rho x_n)] \\ &\quad \times (X^*{}' X^*)^{-1} [X_1^*{}' y_1^* + (\bar{y} - \rho y_n)(\bar{x} - \rho x_n)] \\ &= y_1^*{}' y_1^* - y_1^*{}' X_1^* (X^*{}' X^*)^{-1} X_1^*{}' y_1^* - a^{-1} d^2 \\ &\quad + a(\bar{y} - \rho y_n - a^{-1} d)^2, \end{aligned}$$

where $a = 1 - (\bar{x} - \rho x_n)' (X^*{}' X^*)^{-1} (\bar{x} - \rho x_n)$

and $d = y_1^*{}' X_1^* (X^*{}' X^*)^{-1} (\bar{x} - \rho x_n)$.

Then

$$\begin{aligned} f(\bar{y}, \rho | y) &\propto [y_1^*{}' y_1^* - y_1^*{}' X_1^* (X^*{}' X^*)^{-1} X_1^*{}' y_1^* - a^{-1} d^2 + a(\bar{y} - \rho y_n - a^{-1} d)^2]^{-\frac{(n+1-p)}{2}} \\ &\quad \times |X^*{}' X^*|^{-\frac{1}{2}} \end{aligned}$$

It follows that if ρ is known, then $\bar{y}$ has a t -distribution. Alternatively, we must integrate out $\bar{y}$ in order to find the posterior density of ρ . This is given by

$$(4) \quad f(\rho | y) \propto a^{-\frac{1}{2}} |X^*{}' X^*|^{-\frac{1}{2}} [y_1^*{}' y_1^* - y_1^*{}' X_1^* (X^*{}' X^*)^{-1} X_1^*{}' y_1^* - a^{-1} d^2]^{-\frac{(n-2)}{2}}$$

Numerical integration is used to find the normalising constant for $f(\rho | y)$. This can then be used to estimate some function $g(\bar{y} | y)$, using numerical integration.

4.3 Normal-gamma prior

Assuming a normal-gamma prior density for $f(\beta, \sigma)$ and a beta prior for ρ^2 as before, then the joint posterior density is

$$f(\bar{y}, \beta, \rho, \sigma | y) \propto \sigma^{-(n-p+\bar{v}+2)} \exp \left\{ -\frac{1}{2\sigma^2} [\bar{y}\bar{y}' + (y^* - X^*\beta)'(y^* - X^*\beta) + (\beta - \bar{\beta})'A(\beta - \bar{\beta})] \right\}.$$

After integrating out σ the density becomes

$$\begin{aligned} f(\bar{y}, \beta, \rho | y) &\propto [\bar{y}\bar{y}' + (y^* - X^*\beta)'(y^* - X^*\beta) + (\beta - \bar{\beta})'A(\beta - \bar{\beta})]^{-\frac{(n+p+\bar{v}+1)}{2}} \\ &\propto [\bar{y}\bar{y}' + y^{*'}y^* + \bar{\beta}'A\bar{\beta} - (A\bar{\beta} + X^{*'}y^*)'(X^{*'}X^* + A)^{-1}(A\bar{\beta} + X^{*'}y^*) \\ &\quad + (\beta - (X^{*'}X^* + A)^{-1}(A\bar{\beta} + X^{*'}y^*))'(X^{*'}X^* + A) \\ &\quad \times (\beta - (X^{*'}X^* + A)^{-1}(A\bar{\beta} + X^{*'}y^*))]^{-\frac{(n+p+\bar{v}+1)}{2}} \end{aligned}$$

This is the kernel of a multivariate t -distribution, and on integrating out β , we obtain

$$\begin{aligned} f(\bar{y}, \rho | y) &\propto [\bar{y}\bar{y}' + y^{*'}y^* + \bar{\beta}'A\bar{\beta} - (A\bar{\beta} + X^{*'}y^*)'(X^{*'}X^* + A)^{-1} \\ &\quad \times (A\bar{\beta} + X^{*'}y^*)]^{-\frac{(n+\bar{v}+1)}{2}} |X^{*'}X^* + A|^{-\frac{1}{2}}. \end{aligned}$$

Let $a = 1 - (\bar{z}' - \rho x_n')(X^{*'}X^* + A)^{-1}(\bar{z} - \rho x_n)$, as before. Then

$$\begin{aligned} &y^{*'}y^* + \bar{\beta}'A\bar{\beta} - (A\bar{\beta} + X^{*'}y^*)'(X^{*'}X^* + A)^{-1}(A\bar{\beta} + X^{*'}y^*) \\ &= y_1'y_1 + \bar{\beta}'A\bar{\beta} - (A\bar{\beta} + X_1'y_1)'(X^{*'}X^* + A)^{-1}(A\bar{\beta} + X_1'y_1) \\ &= a^{-1}[(y_1'X_1 + \bar{\beta}'A)(X^{*'}X^* + A)^{-1}(\bar{z} - \rho x_n)]^2 \\ &\quad + a[\bar{y} - \rho y_n - a^{-1}(A\bar{\beta} + X^{*'}y^*)'(X^{*'}X^* + A)^{-1}(\bar{z} - \rho x_n)]^2. \end{aligned}$$

Thus

$$\begin{aligned}
 f(\tilde{y}, \rho | y) &\propto |X^*{}'X^* + A|^{-\frac{1}{2}} \left[v s^2 + y_1^*{}' y_1^* + \tilde{\beta}' A \tilde{\beta} \right. \\
 &\quad - (A \tilde{\beta} + X_1^*{}' y_1^*)' (X^*{}'X^* + A)^{-1} (A \tilde{\beta} + X_1^*{}' y_1^*) \\
 &\quad \left. - a^{-1} [(y_1^*{}' X_1^* + \tilde{\beta}' A) (X^*{}'X^* + A)^{-1} (\bar{z} - \rho x_n)]^2 \right]^{-\frac{(n+\tilde{v})}{2}} \\
 &\quad + a [\tilde{y} - \rho y_n - a^{-1} (A \tilde{\beta} + X_1^*{}' y_1^*)' (X^*{}'X^* + A)^{-1} (\bar{z} - \rho x_n)]^2 \right]^{-\frac{(n+\tilde{v})}{2}}
 \end{aligned}$$

Again it follows that if ρ is known, then $\tilde{y}$ has a t -distribution. As before we can integrate out $\tilde{y}$ to find the density of ρ .

$$\begin{aligned}
 f(\rho | y) &\propto a^{-\frac{1}{2}} |X^*{}'X^* + A|^{-\frac{1}{2}} \left[v s^2 + y_1^*{}' y_1^* + \tilde{\beta}' A \tilde{\beta} \right. \\
 &\quad - (A \tilde{\beta} + X_1^*{}' y_1^*)' (X^*{}'X^* + A)^{-1} (A \tilde{\beta} + X_1^*{}' y_1^*) \\
 &\quad \left. - a^{-1} [(y_1^*{}' X_1^* + \tilde{\beta}' A) (X^*{}'X^* + A)^{-1} (\bar{z} - \rho x_n)]^2 \right]^{-\frac{(n+\tilde{v})}{2}}
 \end{aligned}$$

Numerical integration is used to find the normalising constant for $f(\rho | y)$, and the result can be used to estimate some function of $\tilde{y}$.

5. SIMULATION STUDY

We consider the predictive density of $\tilde{y}$ with a normal-gamma prior and investigate the influence of the hyperparameters v and s^2 on the 95% credibility interval for $\tilde{y}$. We use data generated by Zellner (1971, p. 89), based on the model

$$y_i = 3x_i + u_i$$

$$u_i = \rho u_{i-1} + \epsilon_i, \quad i = 1, 2, \dots, 15$$

with the ϵ_i 's being standard random normal deviates. Zellner (1971) generated two series of y_i 's, with ρ values of 0.5 and 1.25. We generated two further series with ρ values of 0.25 and 0.75, to achieve a wider spread of ρ values.

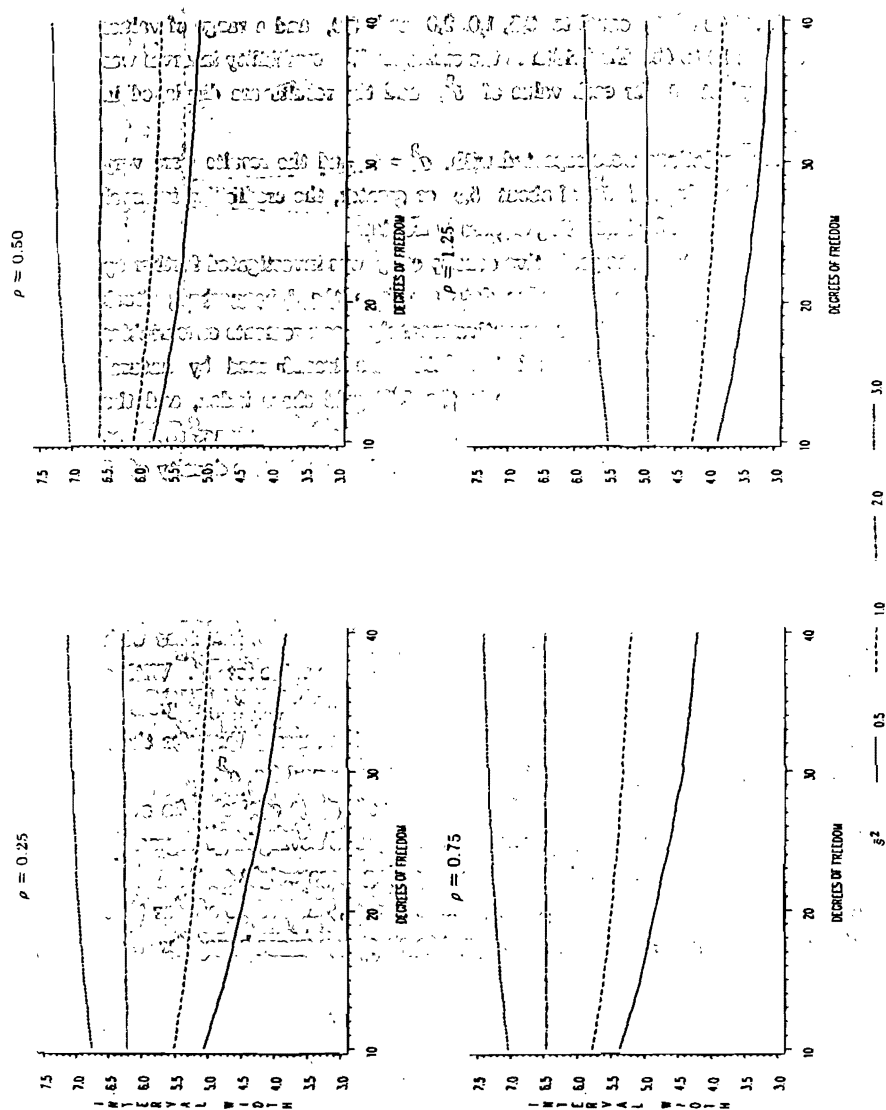


Figure 1. Width of credibility interval for simulated data.

For each series, we computed the width of the 95% credibility interval for $\tilde{y}$, for different values of ν and s^2 . For the other priors it was natural to take $\beta_0 = 0$ and $\beta_1 = 3$, with A as the identity matrix. Since $\sigma^2 = 1$, we took values of s^2 equal to 0.5, 1.0, 2.0 and 3.0, and a range of values for ν from 10 to 40. The width of the corresponding credibility interval was plotted against ν for each value of s^2 , and the results are displayed in figure 1.

The simulations were repeated with $\sigma^2 = 4$, and the results were very similar. For values of s^2 of about 5.5 or greater, the credibility interval increased in width for increasing degrees of freedom.

This property of the predictive density of $\tilde{y}$ was investigated further by considering a real data set. The data are from the Johannesburg Stock Exchange (JSE) and are 100 consecutive monthly measurements on variables relating to gold shares, and all the data were transformed by natural logarithms. The response variable is the JSE gold share index, and the predictor variables are the gold price in rands, the bank acceptance rate, the Dow Jones index and the gold price in dollars. The predictive density of $\tilde{y}$ with a normal-gamma prior was obtained using prior information with $s^2 = 0.05355$ and values 20, 40, 60 and 80 for degrees of freedom. For the purpose of the simulation, the value of s^2 was taken from an ordinary least squares analysis of previous data.

The width of the credibility interval shown in figure 2 increases with increasing degrees of freedom, suggesting an over-large value for s^2 . With a value of $s^2 = 0.015$, the interval width decreases with increasing degrees of freedom, whilst for $s^2 = 0.25$ the interval width increases. Thus, for these data, the simulation suggests that 0.25 is an upper bound for σ^2 .

The results indicate clearly that the value of s^2 is critical. An over-large value of s^2 leads to a wider interval for $\tilde{y}$, and having large degrees of freedom for s^2 exacerbates the problem by giving more weight to this value. One has to be cautious in generalizing these results, but it does appear that the credibility interval for $\tilde{y}$ may be used to obtain an upper bound for σ^2 .

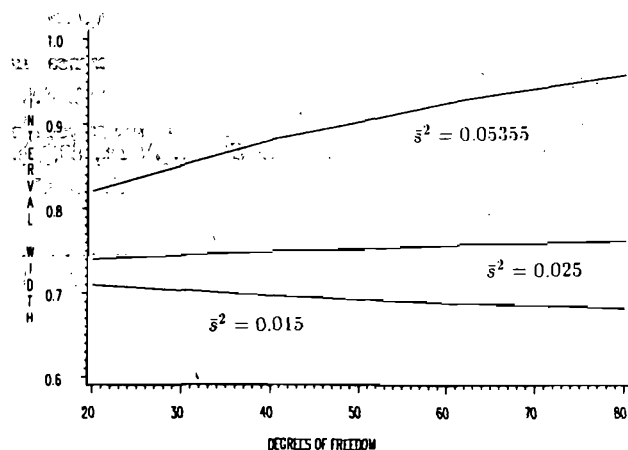


Figure 2. Width of credibility interval for Johannesburg Stock Exchange data.

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